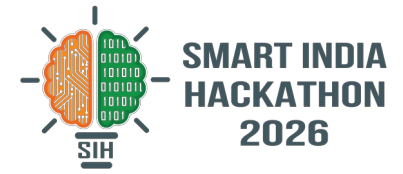


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26171
- **Problem Statement Title** – On-device Visual Perception for Light-weight Browser Agents
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** – _____
- **Team Name (Registered on portal)** – NEXORA

AEGISBROWSE

Privacy-Preserving Visual Browser Agent

"See less. Reveal less. Still complete the task."

 **Video walkthrough:** *paste YouTube link here*



■ PROPOSED SOLUTION — HOW IT ADDRESSES THE PROBLEM

Conventional cloud agent

Raw screen → cloud VLM

✗ Entire private screen is transmitted

Fully local agent

Raw screen → large local pipeline

✗ Constrained by client compute, memory and model capability

AegisBrowse — hybridScreen → local perception → privacy fusion
→ sanitized context → server planner

✓ Server intelligence, no raw-screen exposure

■ WHAT LEAVES THE DEVICE — PRELIMINARY PRIVACY VALIDATION

1 • RAW SCREEN

Transport Department eSeva

Applicant Name	Rohit Khan	Permit Status	APPROVED
Licence Number	MH29 20251234		
Vehicle Number	TN20 ZS 4814		
Correspondence will be sent to the address below. 233, Gandhi Marg, Kochi - 440369			
Login Password	*****		
Download Permit		Print	

local only

2 • LOCAL PERCEPTION

Transport Department eSeva

Applicant Name	Rohit Khan	Permit Status	APPROVED
Licence Number	MH29 20251234		
Vehicle Number	TN20 ZS 4814		
Correspondence will be sent to the address below. 233, Gandhi Marg, Kochi - 440369			
Login Password	*****		
Download Permit		Print	

UI + sensitive regions detected

3 • SERVER CONTEXT

Transport Department eSeva

Applicant Name	[PERSON_1]	Permit Status	APPROVED
Licence Number	[LICENCE_1]		
Vehicle Number	[VEHICLE_1]		
Correspondence will be sent to the address below. [ADDRESS_1]			
Login Password	*****		
Download Permit		Print	

sensitive regions redacted locally

■ INNOVATION AND UNIQUENESS

Multimodal local perception

vision and the DOM cover each other's blind spots

Multimodal privacy fusion

structural + pixel signals build one redaction map

Task-aware sanitized context

goal relevance reduces visual and semantic disclosure

Preliminary held-out tests show the core trade-off: visual escalation improves privacy recall but can over-mask safe context — motivating selective escalation.

■ FINAL ARCHITECTURE — PERCEPTION AND PRIVACY RESOLVE LOCALLY

NETWORK BOUNDARY

ON DEVICE — INSIDE THE BROWSER

1 Screen capture + DOM / accessibility tree

2 Compact visual detector

quantized UI + visual-PII model, WebGPU accelerated

3 Structural signals

roles, labels, field types, input semantics

4 Multimodal screen map

one page model: geometry · semantics · actionability · sensitivity

5 Multimodal privacy fusion

no single detector is trusted — signals vote on one redaction map

6 Task-aware sanitized context

goal-relevance scoring selects the region and metadata actually needed

Cheap structural signals first → expensive OCR / vision only when uncertainty or task need requires it.

ON-DEVICE DETECTOR — TARGET OUTPUT

IN: viewport screenshot + DOM snapshot

button	→	bbox + confidence
input	→	bbox + confidence
text_region	→	bbox + confidence
face	→	bbox + confidence

Exported to ONNX · WebGPU acceleration · WebAssembly fallback · model family selected by benchmark, not assumption.

MULTIMODAL PRIVACY FUSION

TEXT + STRUCTURE

DOM tags · regex
checksums · NER

PIXELS

OCR · faces
visual IDs · QR

ONE REDACTION MAP — applied to pixels and to disclosed metadata

SERVER — open-weight VLM, sanitized context only → strict JSON action

VALIDATE

EXECUTE

VERIFY

■ TECHNOLOGIES

● Browser

Chrome MV3 · Firefox WebExtensions

● Inference

ONNX Runtime Web · WebGPU / WASM · INT8

● Privacy

Regex/checksum · NER · OCR · visual PII

● Server

FastAPI · open-weight VLM · strict JSON

OBSERVE → PERCEIVE → SANITIZE → SELECT CONTEXT → PLAN → VALIDATE → EXECUTE → VERIFY

■ DESIGNED DIRECTLY AROUND THE FIVE PUBLISHED EVALUATION CRITERIA

25%	20%	20%	20%	15%
VISUAL CONTEXT UI / sensitive-region detection Measure: P / R / F1 · IoU	PII DETECTION per-class sensitive-data recognition Measure: Precision / Recall / F1	REDACTION protect privacy without destroying context Measure: pixel & span P / R	CLIENT RESOURCES lightweight browser execution Measure: RAM · model · WebGPU / WASM	END-TO-END LATENCY observe → verified action Measure: stage-wise ms

■ PRELIMINARY VALIDATION — REFERENCE BENCHMARK, 48 HELD-OUT PAGES

Perception pipeline	PII precision	PII recall	Redaction recall	Redaction precision
Structural signals only	100.0%	68.5%	63.2%	100.0%
+ contextual entity detection	100.0%	83.8%	71.7%	100.0%
+ visual / OCR escalation	91.8%	96.8%	98.4% ↑	55.9% ↓

Why selective escalation?

Vision closes privacy blind spots — but always-on vision over-masks safe content.

■ HOW THE FINAL SYSTEM WILL BE PROVEN

4 Baselines	Held-out Domains	Layer Ablation	Multi-device Profiling
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■ POTENTIAL IMPACT ON THE TARGET AUDIENCE

Organizations and developers deploying AI agents over authenticated, privacy-sensitive browser workflows

They face a hard trade-off between capable server-side reasoning and exposing sensitive visual context. AegisBrowse is designed to reduce that trade-off through local perception and controlled disclosure.

EXAMPLE DEPLOYMENT DOMAINS

Government services

certificates, grievances,
benefit applications

Education

scholarships, admissions,
examination portals

Banking & finance

KYC-heavy account and
statement screens

Healthcare

patient records and
appointment portals

Human resources

onboarding, payroll,
employee self-service

■ BENEFITS OF THE SOLUTION

Social

PRIVACY

Sensitive identity context processed locally,
before remote reasoning.

Economic

EFFICIENCY

Less sensitive context and visual payload
transmitted per automated task.

Operational

DEPLOYMENT

Compatible open-weight planners run on
organization-controlled infrastructure.

Technical

REUSABILITY

A browser privacy layer applicable across
authenticated workflows.

■ HOW IMPACT WILL BE EVIDENCED — MEASURED, NEVER ASSERTED

Task success rate

Privacy leakage

Context exposure

Client resource cost

End-to-end latency

Server-grade intelligence on private screens — without making the raw screen the default server input.

■ PRIOR WORK → THE GAP → OUR POSITION

PRIOR WORK

Raw screenshot → cloud VLM

the entire visual context leaves the device

DOM / accessibility only

misses canvas, images, scanned documents

Fully redacted screenshot

still ships unnecessary safe context every step

Fully local agent

bounded by client compute, memory and model capability

THE GAP

Existing approaches separately optimize perception, privacy, or agent capability.

The unresolved question is how to jointly minimize disclosure while preserving task success under browser compute constraints.

Public benchmarks give limited coverage of Indian government-portal PII patterns, so we build and release our own.

OUR POSITION

Multimodal privacy fusion

no single detector is trusted with all sensitive data

Task-aware sanitized context

the screen stops being the unit of disclosure

Selective cascaded perception

expensive vision runs only where it is needed

Measured, not asserted

five official metrics, four baselines, open benchmark

AegisBrowse does not treat sanitization as a single redaction step: it fuses structural and visual privacy signals, escalates perception selectively, and transmits task-relevant sanitized context instead of the whole screen.

■ DETAILS / LINKS OF REFERENCE AND RESEARCH WORK

- Deng et al. — Mind2Web: Towards a Generalist Agent for the Web · arXiv 2306.06070
- Zheng et al. — SeeAct: GPT-4V(ision) is a Generalist Web Agent, if Grounded · arXiv 2401.01614
- Lu et al. — OmniParser for Pure Vision Based GUI Agent · arXiv 2408.00203
- Zhang et al. — PrivWeb: Content-aware Privacy Protection for Web Agents · arXiv 2509.11939
- Zhao — WebPII: Benchmarking Visual PII Detection · arXiv 2603.17357

SUPPORTING MATERIAL

Full literature review, benchmark and implementation available in supporting material.